

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	SECTION:	CPE31S2
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ACTIVITY NO. 10.1 Install, Configure and Manage Enterprise Log Monitoring via Ansible

1. PROCEDURE OUTPUT

I. Import ElasticSearch GPG Key

1. Create the roles directory for ElasticSearch with `mkdir -p roles/ElasticSearch/tasks`
2. Create the role playbook with `nano roles/ElasticSearch/tasks/main.yml` and create the play for gpg key installation.

```
qimagalan-workstation@workstation: ~/... x qimagalan-workstation@workstation: ~/E... x qimagalan-workstation@workstation: ~/E...
GNU nano 7.2 roles/ElasticSearch/tasks/main.yml

- name: Create directory for installation
  ansible.builtin.file:
    path: "{{ elastic_dir }}"
    state: directory

- name: Download the signing key
  become: true
  ansible.builtin.get_url:
    url: "{{ signkey_url }}"
    dest: "{{ elastic_dir }}"
  when: ansible_distribution == "Ubuntu"

- name: Extract gpg key
  become: true
  ansible.builtin.command:
    cmd: gpg --dearmor --output /usr/share/keyrings/elasticsearch-keyring.gpg "{{ elastic_dir }}"
    creates: /usr/share/keyrings/elasticsearch-keyring.gpg
  when: ansible_distribution == "Ubuntu"

- name: Import Signing Key for CentOS
  become: true
  ansible.builtin.rpm_key:
    key: "{{ signkey_url }}"
    state: present
  when: ansible_distribution == "CentOS"
```

This creates the directory we need to work with the ElasticStack plus it downloads the signing key as preparatory step as recommended in the documentation:

<https://www.elastic.co/docs/deploy-manage/deploy/self-managed/install-elasticsearch-with-debian-package#deb-key>

3. Create the following variables inside `roles/ElasticSearch/vars/main.yml`

```
qimagalan-workstation@workstation: ~/... x qimagalan-workstation@workstation: ~/E... x qimagalan-workstation@workstation: ~/E... x
GNU nano 7.2 roles/ElasticSearch/vars/main.yml
---
signkey_url: "https://artifacts.elastic.co/GPG-KEY-elasticsearch"
debpkg_link: "https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.5.4-amd64.deb"
debpkg_chksum: "https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.5.4-amd64.deb.sha512"
rpmkg_link: "https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.5.4-x86_64.rpm"
rpmkg_chksum: "https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.5.4-x86_64.rpm.sha512"
elastic_dir: "{{ ansible_env.HOME }}/ElasticStack"
```

4. Run the playbook to install the gpg keys for Ubuntu and CentOS

```
qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan$ ansible-playbook install_elasticstack.yml -i
inventory.yml -K
BECOME password:

PLAY [web_servers,file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.114]
ok: [192.168.56.113]

TASK [ElasticSearch : Create directory for installation] *****
ok: [192.168.56.113]
ok: [192.168.56.114]

TASK [ElasticSearch : Download the signing key] *****
skipping: [192.168.56.114]
ok: [192.168.56.113]

TASK [ElasticSearch : Extract gpg key] *****
skipping: [192.168.56.114]
ok: [192.168.56.113]

TASK [ElasticSearch : Import Signing Key for CentOS] *****
skipping: [192.168.56.113]
ok: [192.168.56.114]

PLAY RECAP *****
192.168.56.113      : ok=4    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    igno
red=0
192.168.56.114      : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    igno
red=0
```

```
qimagalan-workstation@server1:~$ ls ~/ElasticStack
GPG-KEY-elasticsearch
qimagalan-workstation@server1:~$
```

II. Download from the debian and rpm links

1. Modify the playbook and the play that will install the deb and rpm packages

```
- name: Download Elastic Search deb pkg
  ansible.builtin.get_url:
    url: "{{ debpkg_link }}"
    dest: "{{ elastic_dir }}"
    checksum: "sha512:{{ debpkg_chksum }}"
    when: ansible_distribution == "Ubuntu"

- name: Download Elastic Search rpm pkg
  ansible.builtin.get_url:
    url: "{{ rpmpkg_link }}"
    dest: "{{ elastic_dir }}"
    checksum: "sha512:{{ rpmpkg_chksum }}"
    when: ansible_distribution == "CentOS"
```

2. Run the playbook and check results

```
TASK [ElasticSearch : Extract gpg key] *****
skipping: [192.168.56.114]
ok: [192.168.56.113]

TASK [ElasticSearch : Import Signing Key for CentOS] *****
skipping: [192.168.56.113]
ok: [192.168.56.114]

TASK [ElasticSearch : Download Elastic Search deb pkg] *****
skipping: [192.168.56.114]
changed: [192.168.56.113]

TASK [ElasticSearch : Download Elastic Search rpm pkg] *****
skipping: [192.168.56.113]
changed: [192.168.56.114]

PLAY RECAP *****
192.168.56.113      : ok=5    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    igno
red=0
192.168.56.114      : ok=4    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    igno
red=0

qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan$
```

```
qimagalan-workstation@server1:~$ ls ~/ElasticStack
elasticsearch-9.5.4-amd64.deb  GPG-KEY-elasticsearch
qimagalan-workstation@server1:~$
```

III. Install the packages downloaded

1. Add these variables to the roles/ElasticSearch/vars/main.yml

```
qimagalan@localhost:~
x qimagalan-workstation@workstation: ~/E...
x qimagalan-workstation@workstation: ~/E...
GNU nano 7.2 roles/ElasticSearch/vars/main.yml *
---
signkey_url: "https://artifacts.elastic.co/GPG-KEY-elasticsearch"
debpkg_link: "https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.5.4-amd64.deb"
debpkg_chksum: "https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.5.4-amd64.deb.sha512"
rpmkg_link: "https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.5.4-x86_64.rpm"
rpmkg_chksum: "https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.5.4-x86_64.rpm.sha512"
elastic_dir: "{{ ansible_env.HOME }}/ElasticStack"

deb_elastic: "{{ elastic_dir }}/elasticsearch-9.5.4-amd64.deb"
rpm_elastic: "{{ elastic_dir }}/elasticsearch-9.5.4-x86_64.rpm"
```

This declares all necessary variables towards the links of the package of the elasticsearch, and it installs them on the `elastic_dir` which could easily be changed by simply modifying this variable file.

2. Add this to the bottom of the playbook to install from packages

```
- name: Install from deb pkg
  tags: install
  become: true
  ansible.builtin.apt:
    deb: "{{ deb_elastic }}"
  when: ansible_distribution == "Ubuntu"

- name: Install from rpm pkg
  tags: install
  become: true
  ansible.builtin.dnf:
    name: "{{ rpm_elastic }}"
    state: present
  when: ansible_distribution == "CentOS"
```

[^]G Help [^]O Write Out [^]W Where Is [^]K Cut [^]T Execute [^]C Location M-U Undo
[^]X Exit [^]R Read File [^]\ Replace [^]U Paste [^]J Justify [^]/ Go To Line M-E Redo

3. Run the playbook and check results

```
qimagalan-workstation@workstation: ~/... x qimagalan-workstation@workstation: ~/E... x qimagalan-workstation@workstation: ~/E... x
qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan$ ansible-playbook install_elasticstack.yml -i
inventory.yml -K --tags "install"
BECOME password:

PLAY [web_servers,file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.113]
ok: [192.168.56.114]

TASK [ElasticSearch : Install from deb pkg] *****
skipping: [192.168.56.114]
changed: [192.168.56.113]

TASK [ElasticSearch : Install from rpm pkg] *****
skipping: [192.168.56.113]
changed: [192.168.56.114]

PLAY RECAP *****
192.168.56.113      : ok=2    changed=1    unreachable=0    failed=0    skipped=1    rescued=0    igno
red=0
192.168.56.114      : ok=2    changed=1    unreachable=0    failed=0    skipped=1    rescued=0    igno
red=0
qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan$
```

```
qimagalan@localhost:~ x qimagalan-workstation@workstation: ~/E... x qimagalan-workstation@workstation: ~/E... x
qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan$ ssh qimagalan@Galan_CentOS
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Tue Sep 29 17:11:36 2026 from 192.168.56.1
[qimagalan@localhost ~]$ sudo systemctl status elasticsearch
[sudo] password for qimagalan:
○ elasticsearch.service - Elasticsearch
   Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: https://www.elastic.co
[qimagalan@localhost ~]$
```

IV. Setup Nodes for connection

1. Add another play at the bottom to place basic configuration on the hosts

```
- name: Configure the elasticsearch.yml for both hosts
  tags: config
  become: true
  ansible.builtin.blockinfile:
    path: /etc/elasticsearch/elasticsearch.yml
    block: |
      cluster.name: my-cluster
      network.host: 0.0.0.0
      transport.host: 0.0.0.0
    marker: "# {mark} ANSIBLE MANAGED ELASTICSEARCH CONFIG"
```

^G Help

^X Exit

^O Write Out

^R Read File

^W Where Is

^_ Replace

^K Cut

^U Paste

^T Execute

^J Justify

The configuration above is the configuration recommended in the elasticsearch documentation. To manage it blockinfile is used with markers. In future modifications, we only have to specify the marker in order to change the contents of a block.

2. Run and check if configurations reflected

```
PLAY [web_servers,file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.113]
ok: [192.168.56.114]

TASK [ElasticSearch : Configure the elasticsearch.yml for both hosts] *****
changed: [192.168.56.113]
changed: [192.168.56.114]

PLAY RECAP *****
192.168.56.113      : ok=2    changed=1    unreachable=0    failed=0    skipped=
0    rescued=0    ignored=0
192.168.56.114      : ok=2    changed=1    unreachable=0    failed=0    skipped=
0    rescued=0    ignored=0

qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan$
```

```
#transport.host: 0.0.0.0

#----- END SECURITY AUTO CONFIGURATION -----
# BEGIN ANSIBLE MANAGED ELASTICSEARCH CONFIG
cluster.name: my-cluster
network.host: 0.0.0.0
transport.host: 0.0.0.0
# END ANSIBLE MANAGED ELASTICSEARCH CONFIG
qimagalan-workstation@server1:~$
```

3. Add the play to start and enable the elasticsearch service

```
- name: Start the elasticsearch service
  tags: start
  become: true
  ansible.builtin.service:
    name: elasticsearch
    state: started
    enabled: true

[ Wrote 64 lines ]

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Exe
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Jus
```

4. Run playbook to start service first

```
PLAY [web_servers,file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.113]
ok: [192.168.56.114]

TASK [ElasticSearch : Start the elasticsearch service] *****
changed: [192.168.56.114]
changed: [192.168.56.113]

PLAY RECAP *****
192.168.56.113      : ok=2    changed=1    unreachable=0    failed=0    skipped=
0    rescued=0    ignored=0
192.168.56.114      : ok=2    changed=1    unreachable=0    failed=0    skipped=
0    rescued=0    ignored=0
qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan$
```

```
qimagalan-workstation@server1:~$ sudo systemctl status elasticsearch
[sudo] password for qimagalan-workstation:
● elasticsearch.service - Elasticsearch
   Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; enabled; preset: e>
   Active: active (running) since Tue 2026-09-29 17:21:59 PST; 1min 8s ago
     Docs: https://www.elastic.co
   Main PID: 11271 (server-launcher)
    Tasks: 104 (limit: 1432)
   Memory: 1.2G (peak: 1.3G swap: 446.8M swap peak: 446.8M)
      CPU: 3min 5.431s
   CGroup: /system.slice/elasticsearch.service
           └─11271 /usr/share/elasticsearch/lib/tools/server-launcher/server-launcher>
             └─11327 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.cache.t>
               └─11354 /usr/share/elasticsearch/modules/x-pack-ml/platform/linux-x86_64/>

Sep 29 17:19:49 server1 systemd[1]: Starting elasticsearch.service - Elasticsearch...
Sep 29 17:20:18 server1 systemd-entrypoint[11271]: WARNING: Unknown module: org.apache>
Sep 29 17:21:54 server1 systemd-entrypoint[11271]: WARNING: A topically unsupported m>
```

5. Reload the system daemon as well by adding a new play

```
- name: Reload systemd daemon
  tags: reload
  become: true
  ansible.builtin.systemd:
    daemon_reload: true
```

```
qimagalan-workstation@serv... x qimagalan-workstation@work... x qimagalan-workstation@work... x
ll_elasticsearch.yml -i inventory.yml -K --tags "reload"
BECOME password:

PLAY [web_servers,file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.113]
ok: [192.168.56.114]

TASK [ElasticSearch : Reload systemd daemon] *****
ok: [192.168.56.113]
ok: [192.168.56.114]

PLAY RECAP *****
192.168.56.113 : ok=2 changed=0 unreachable=0 failed=0 skipped=
0 rescued=0 ignored=0
192.168.56.114 : ok=2 changed=0 unreachable=0 failed=0 skipped=
0 rescued=0 ignored=0
```

6. Then reset the password by doing:

`/usr/share/elasticsearch/bin/elasticsearch-reset-password -u elastic`

7. Take note of the password as it will be used to verify the installation of elasticsearch

```
qimagalan@localhost:/usr/share/elasticsearch/bin x qimagalan-workstation@v
[qimagalan@localhost bin]$ sudo ./elasticsearch-reset-password -u elastic
[sudo] password for qimagalan:
This tool will reset the password of the [elastic] user to an autogenerated value.
The password will be printed in the console.
Please confirm that you would like to continue [y/N]y

Password for the [elastic] user successfully reset.
New value: GPWqTa9+v9ag*z7sMTsD
[qimagalan@localhost bin]$
```

Password resetting should always be done manual since elasticsearch would give a new random password that needs to be kept.

V. Verify Installation of Elastic Search

1. Run `sudo curl --cacert /etc/elasticsearch/certs/http_ca.crt -u elastic https://localhost:9200`. Then type the password from the reset earlier.

```
qimagalan-workstation@server1:~$ sudo curl --cacert /etc/elasticsearch/certs/http_ca.crt \
-u elastic https://localhost:9200
Enter host password for user 'elastic':
{
  "name" : "server1",
  "cluster_name" : "my-cluster",
  "cluster_uuid" : "7CVA-1HmTaG2Tu204WPK1Q",
  "version" : {
    "number" : "9.5.4",
    "build_flavor" : "default",
    "build_type" : "deb",
    "build_hash" : "9170df19cae1adb107b7b489b4d82dec66d7a337",
    "build_date" : "2026-09-09T22:42:53.976833287Z",
    "build_snapshot" : false,
    "lucene_version" : "10.5.1",
    "minimum_wire_compatibility_version" : "8.19.0",
    "minimum_index_compatibility_version" : "8.0.0"
  },
  "tagline" : "You Know, for Search"
}
```

Here we verified successfully that the cluster in server1 is running

```
qimagalan@localhost: /usr/share/elasticsearch/bin x qimagalan-workstation
[qimagalan@localhost bin]$ sudo curl --cacert /etc/elasticsearch/certs/http_ca.crt -
Enter host password for user 'elastic':
{
  "name" : "localhost.localdomain",
  "cluster_name" : "my-cluster",
  "cluster_uuid" : "q4gPglA0TfGFkU0i86W4Mw",
  "version" : {
    "number" : "9.5.4",
    "build_flavor" : "default",
    "build_type" : "rpm",
    "build_hash" : "9170df19cae1adb107b7b489b4d82dec66d7a337",
    "build_date" : "2026-09-09T22:42:53.976833287Z",
    "build_snapshot" : false,
    "lucene_version" : "10.5.1",
    "minimum_wire_compatibility_version" : "8.19.0",
    "minimum_index_compatibility_version" : "8.0.0"
  },
  "tagline" : "You Know, for Search"
}
[qimagalan@localhost bin]$
```

Same goes for server 2 the cluster is also running

VI. Install Kibana through deb and rpm package

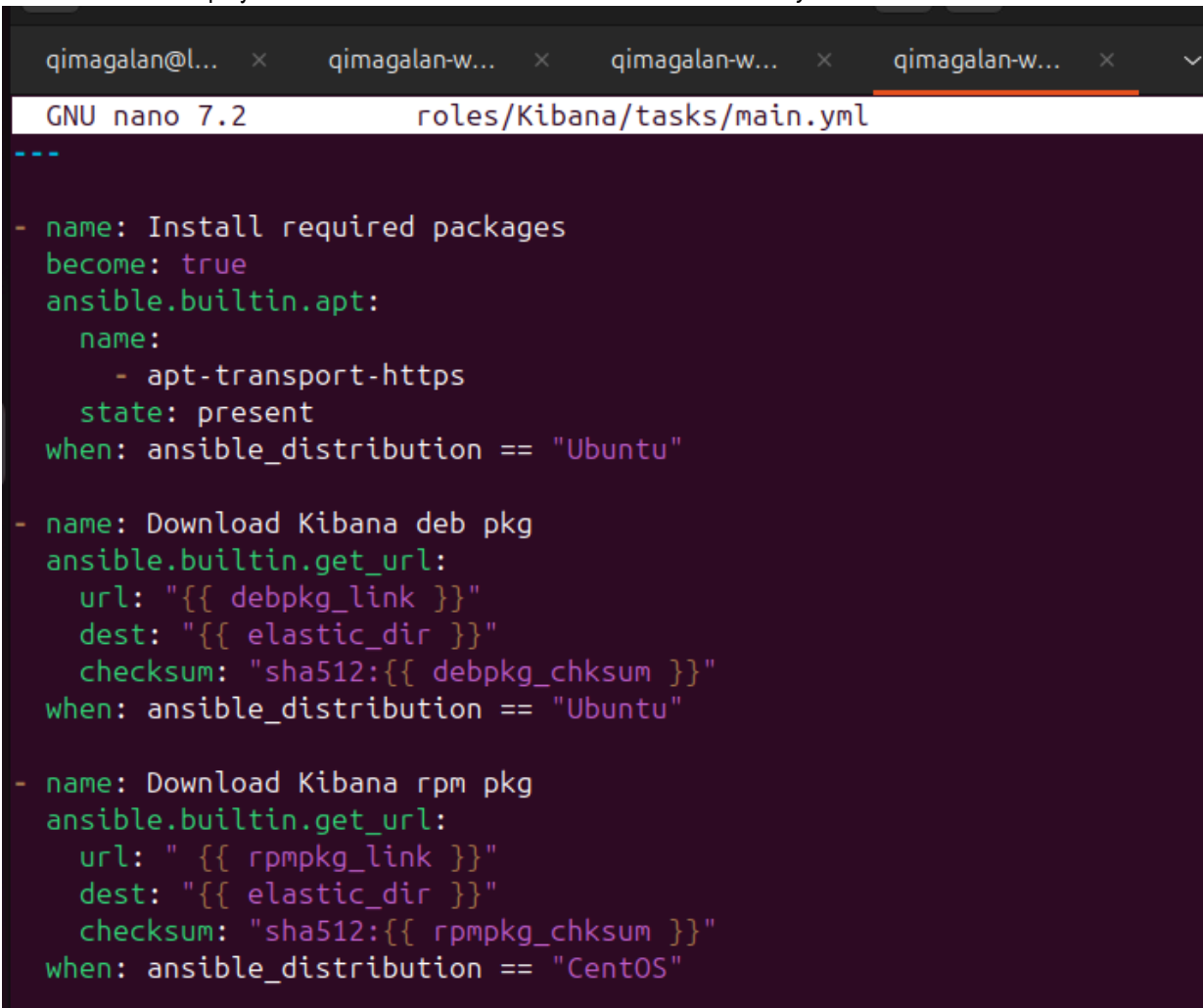
1. Create the role for the Kibana installation: `mkdir -p roles/Kibana/tasks`
2. Create the roles/Kibana/vars/main.yml file and place the following variables

```
qimagalan@localhost: ~ x qimagalan-workstation@server1: ~ x qimagalan-workstation@workstation: ~/Ela...
GNU nano 7.2 roles/Kibana/vars/main.yml
--
debpkg_link: "https://artifacts.elastic.co/downloads/kibana/kibana-9.5.4-amd64.deb"
debpkg_chksum: "https://artifacts.elastic.co/downloads/kibana/kibana-9.5.4-amd64.deb.sha512"
rpmkg_link: "https://artifacts.elastic.co/downloads/kibana/kibana-9.5.4-x86_64.rpm"
rpmkg_chksum: "https://artifacts.elastic.co/downloads/kibana/kibana-9.5.4-x86_64.rpm.sha512"
elastic_dir: "{{ ansible_env.HOME }}/ElasticStack"

deb_elastic: "{{ elastic_dir }}/kibana-9.5.4-amd64.deb"
rpm_elastic: "{{ elastic_dir }}/kibana-9.5.4-x86_64.rpm"
```

We keep the same idea of a vars file above to keep all links and destinations in one place. Honestly, `elastic_dir` might be better off in a `group_vars` since it is repeating for both files.

3. Create the playbook to install Kibana *roles/Kibana/tasks/main.yml*

A screenshot of a terminal window with a dark background. The window title bar shows four tabs, all labeled 'qimagalan-w...'. The active tab is highlighted. Below the title bar, the terminal shows the prompt 'GNU nano 7.2' followed by the file path 'roles/Kibana/tasks/main.yml'. The content of the file is displayed in a light-colored monospace font. It starts with three dashes '---' and contains three Ansible tasks. Each task is preceded by a hyphen and a name. The first task is for installing required packages on Ubuntu. The second task is for downloading Kibana deb packages on Ubuntu. The third task is for downloading Kibana rpm packages on CentOS. The tasks use Jinja2 templating for variables like 'debpkg_link', 'elastic_dir', 'rpmkg_link', and 'rpmkg_chksum'.

```
---  
  
- name: Install required packages  
  become: true  
  ansible.builtin.apt:  
    name:  
      - apt-transport-https  
    state: present  
  when: ansible_distribution == "Ubuntu"  
  
- name: Download Kibana deb pkg  
  ansible.builtin.get_url:  
    url: "{{ debpkg_link }}"  
    dest: "{{ elastic_dir }}"  
    checksum: "sha512:{{ debpkg_chksum }}"  
  when: ansible_distribution == "Ubuntu"  
  
- name: Download Kibana rpm pkg  
  ansible.builtin.get_url:  
    url: "{{ rpmkg_link }}"  
    dest: "{{ elastic_dir }}"  
    checksum: "sha512:{{ rpmkg_chksum }}"  
  when: ansible_distribution == "CentOS"
```

```
- name: Install from deb pkg
tags: install
become: true
ansible.builtin.apt:
  deb: "{{ deb_elastic }}"
when: ansible_distribution == "Ubuntu"

- name: Install from rpm pkg
tags: install
become: true
ansible.builtin.dnf:
  name: "{{ rpm_elastic }}"
  state: present
when: ansible_distribution == "CentOS"
```

4. Run the playbook and verify results

```
qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan$ ansible-playbook ins
lasticstack.yml -i inventory.yml -K --tags "kibana"
BECOME password:

PLAY [web_servers,file_servers] *****

PLAY [web_servers,file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.113]
ok: [192.168.56.114]

TASK [Kibana : Install required packages] *****
skipping: [192.168.56.114]
ok: [192.168.56.113]

TASK [Kibana : Download Kibana deb pkg] *****
skipping: [192.168.56.114]
changed: [192.168.56.113]

TASK [Kibana : Download Kibana rpm pkg] *****
skipping: [192.168.56.113]
ok: [192.168.56.114]

TASK [Kibana : Install from deb pkg] *****
skipping: [192.168.56.114]
changed: [192.168.56.113]

TASK [Kibana : Install from rpm pkg] *****
skipping: [192.168.56.113]
ok: [192.168.56.114]

PLAY RECAP *****
192.168.56.113      : ok=4    changed=2    unreachable=0    failed=0    skippe
rescued=0    ignored=0
```

```
qimagalan@localhost:~ x qimagalan-workstation@server1: ~
[qimagalan@localhost ~]$ ls ~/ElasticStack
elasticsearch-9.5.4-x86_64.rpm kibana-9.5.4-x86_64.rpm
[qimagalan@localhost ~]$
```

5. Generate a kibana enrollment token by adding this play

```
[qimagalan@localhost ElasticStack]$ ls
elasticsearch-9.5.4-x86_64.rpm  kibana-9.5.4-x86_64.rpm  kibanatoken
[qimagalan@localhost ElasticStack]$ cat kibanatoken
"eyJ2ZXIiOiI4LjE0LjAiLCJhZHIIiOlisiMTAucMc4YlJlE10jkyMDAiXSwiZmdyIjoIiYjYjZTU3YWJmZTI0NDJlZTE0NWVjNWElODkyZGU2MzhINDMyMzI4MzJlOTZkZWl5NWZjYzZkM2VhOWYyZm5NiIsImtleSI6IilhzNVQ3S0FCcTY3UjJyYVdhHTkho0mZHS1M3ekhPaXBVeW04ZzlhV3Y5eHcifQ=="
[qimagalan@localhost ElasticStack]$
```

```
qimagalan-workstation@server1:~$ cat ~/ElasticStack/kibanatoken
"eyJ2ZXIiOiI4LjE0LjAiLCJhZHIiOiIsImTAUmc4YlJlE10jkyMDAiXSwiZmlyb3R5IjoiaWJmNzU0Yzc2MzAxMTliMzQ0OGM3MWJkODNiNWZjODk1NDVlNDEzOGM0MzRkZDcxMzljMjY2OWM3Y2Q3NGM3OSIsImtleSI6ImRpRmBm43YUFCQnJBV0pTTjJlak5ROkRWRFo1YVFSWFlqbWljTwd3aWV6V1EifQ=="
```

7. Start and Enable the kibana service by adding this in the Kibana roles tasks

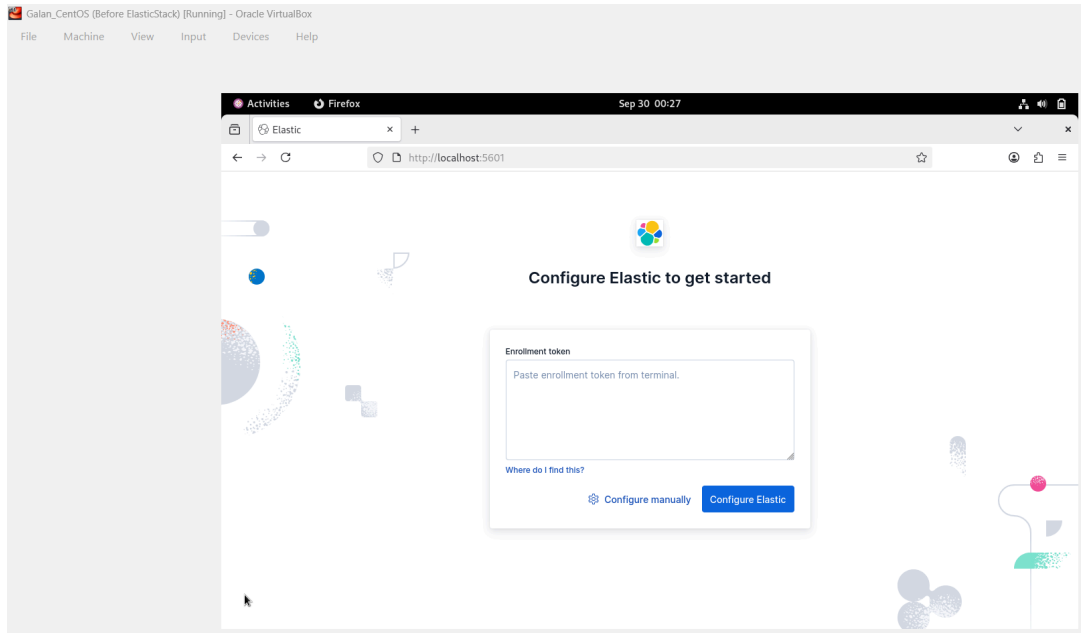
```
- name: Start and Enable Kibana Service
  ansible.builtin.service:
    name: kibana
    state: started
    enabled: true
  become: true

- name: Reload systemd daemon
  tags: reload
  become: true
  ansible.builtin.systemd:
    daemon_reload: true
```

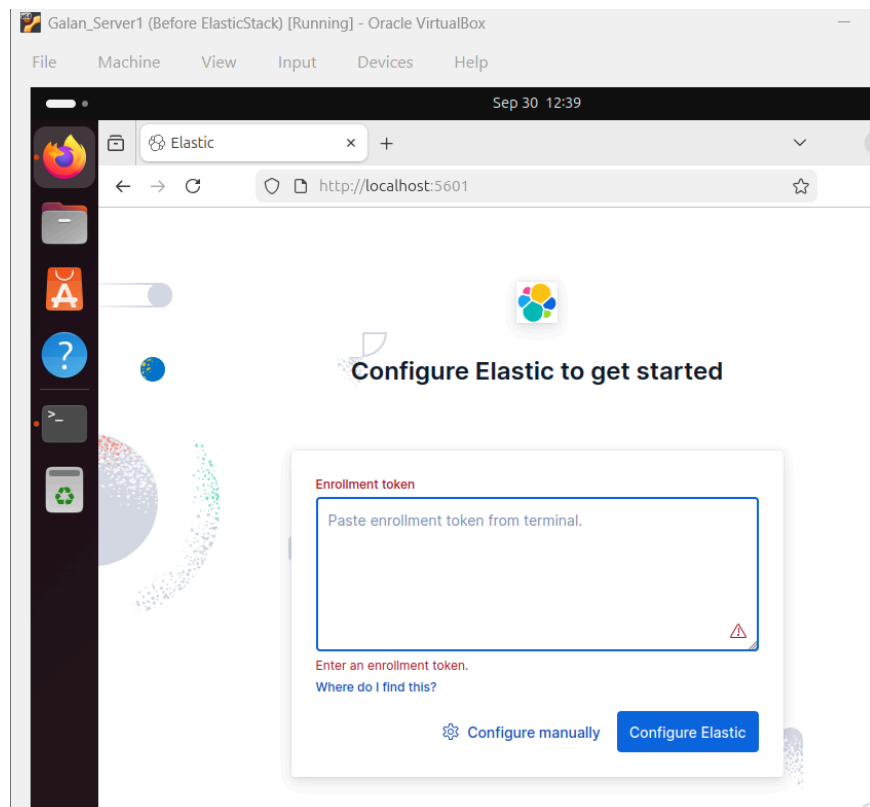
```
[qimagalan@localhost ElasticStack]$ sudo systemctl status kibana
[sudo] password for qimagalan:
● kibana.service - Kibana
   Loaded: loaded (/usr/lib/systemd/system/kibana.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-09-29 16:47:31 PST; 59s ago
     Docs: https://www.elastic.co
   Main PID: 17916 (MainThread)
      Tasks: 11 (limit: 10500)
     Memory: 487.6M (peak: 487.7M)
        CPU: 28.668s
    CGroup: /system.slice/kibana.service
           └─17916 /usr/share/kibana/bin/../../node/default/bin/node /usr/share/kibana/bin/

Sep 29 16:47:31 localhost.localdomain systemd[1]: Started Kibana.
Sep 29 16:47:57 localhost.localdomain kibana[17916]: {"log.level":"info","@timestamp":"2026-09-29T16:47:57.000Z","@version":"8.17.0","service.type":"kibana","log.offset":1}
Sep 29 16:47:57 localhost.localdomain kibana[17916]: Native global console methods have been deprecated. Please use the log API instead.
Sep 29 16:48:11 localhost.localdomain kibana[17916]: [2026-09-29T16:48:11.356+08:00][INFO] Starting Kibana
Sep 29 16:48:11 localhost.localdomain kibana[17916]: [2026-09-29T16:48:11.509+08:00][INFO] Kibana started
lines 1-16/16 (END)
```

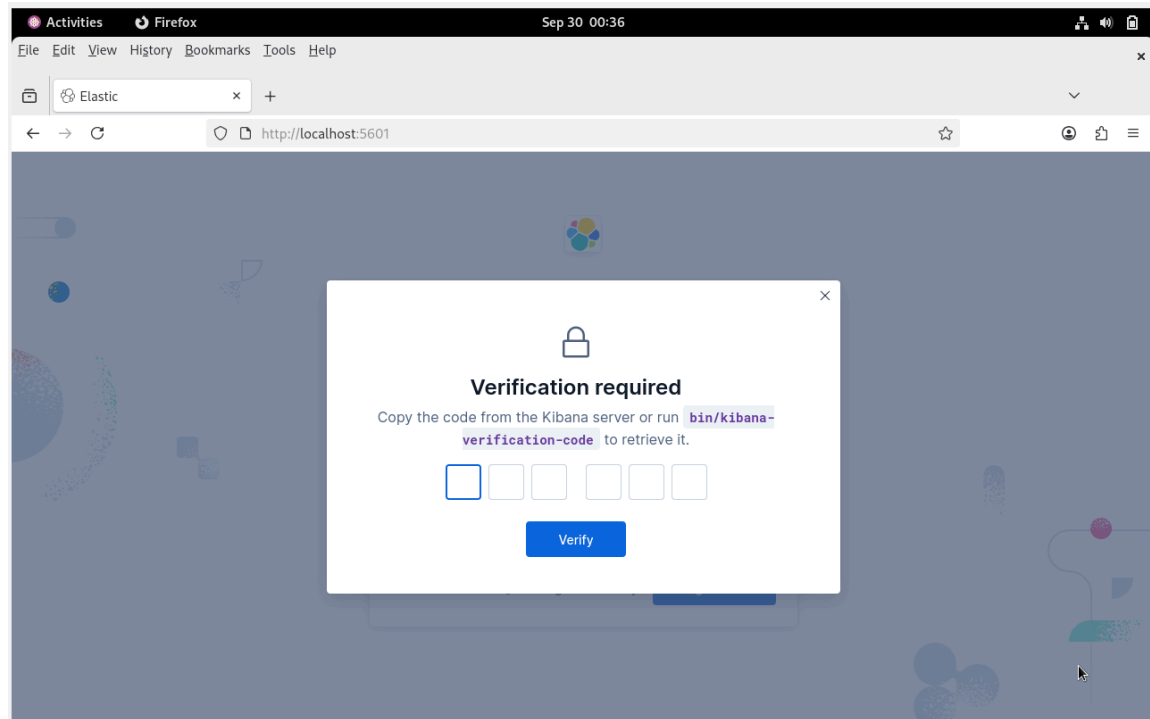
8. Verify the installation of kibana in port 5601
CentOS:



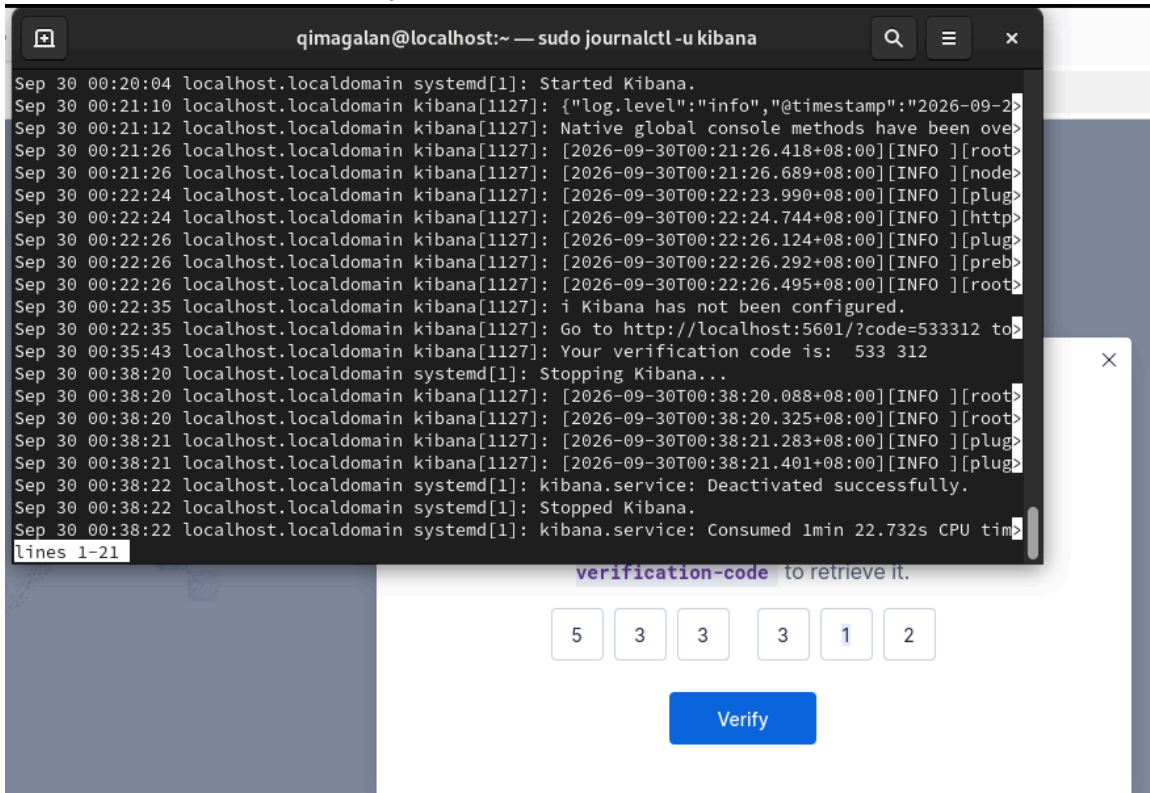
Ubuntu:



9. Use the generated enrollment token from the play in the former steps and paste in the box



10. The code is viewable via `sudo journalctl -u kibana`



The verification code can be seen on the logs stating that it is 533 312

VII. Install Logstash

1. Create a role for logstash through `mkdir -p roles/Logstash/tasks`
2. Create the Logstash play for installing logstash from apt and yum repositories through `nano roles/Logstash/tasks/main.yml`

```
qimagalan-workstation@workstation: ~/ElasticStackInstaller_Galan
qimagalan-workstation@workstation: ~/ElasticStackInstaller_Galan x qimagalan-workstation@workstation: ~/ElasticStackInstaller_Galan x
GNU nano 7.2 roles/Logstash/tasks/main.yml *
--
- name: Add the apt repository to sources list
  become: true
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elastic-keyring.gpg] https://artifacts.elastic.co/packages/9.x/apt stable main"
    filename: elastic-9.x
    state: present
    update_cache: true #refresh the repository list after adding
  when: ansible_distribution == "Ubuntu"

- name: Add the yum repository source
  become: true
  ansible.builtin.yum_repository:
    name: elastic-9.x
    description: Elastic repository for 9.x packages
    baseurl: https://artifacts.elastic.co/packages/9.x/yum
    gpgcheck: true
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: true
    file: elastic-9.x

- name: Install the logstash from repository source
  become: true
  ansible.builtin.package:
    name: logstash
    state: present
```

3. Run playbook and check if Logstash is installed



```
qimagalan-workstation@workstation: ~/ElasticStackInstaller_Galan
qimagalan-workstation@workstation: ~/ElasticSta... x qimagalan-workstation@workstation: ~/ElasticSta... x qimagalan-workstation@server1: ~ x
ok: [192.168.56.114]
ok: [192.168.56.113]

TASK [Logstash : Download the signing key] *****
skipping: [192.168.56.114]
ok: [192.168.56.113]

TASK [Logstash : Extract gpg key] *****
skipping: [192.168.56.114]
ok: [192.168.56.113]

TASK [Logstash : Add the apt repository to sources list] *****
skipping: [192.168.56.114]
ok: [192.168.56.113]

TASK [Logstash : Add the yum repository source] *****
skipping: [192.168.56.113]
ok: [192.168.56.114]

TASK [Logstash : Refresh repository list before installing logstash] *****
changed: [192.168.56.113]
ok: [192.168.56.114]

TASK [Logstash : Install the logstash from repository source] *****
ok: [192.168.56.113]
ok: [192.168.56.114]

PLAY RECAP *****
192.168.56.113      : ok=6   changed=1  unreachable=0  failed=0  skipped=1  rescued=0  ignored=0
192.168.56.114      : ok=4   changed=0  unreachable=0  failed=0  skipped=3  rescued=0  ignored=0

qimagalan-workstation@workstation: ~/ElasticSta... x qimagalan-workstation@workstation: ~/ElasticSta... x
qimagalan-workstation@server1:~$ sudo systemctl status logstash
[sudo] password for qimagalan-workstation:
○ logstash.service - logstash
   Loaded: loaded (/usr/lib/systemd/system/logstash.service; disabled; preset: enabled)
   Active: inactive (dead)

qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan$ ssh qimagalan@Galan_CentOS
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Wed Sep 30 10:46:12 2026 from 192.168.56.1
[qimagalan@localhost ~]$ sudo systemctl status logstash
[sudo] password for qimagalan:
○ logstash.service - logstash
   Loaded: loaded (/usr/lib/systemd/system/logstash.service; disabled; preset: disabled)
   Active: inactive (dead)
[qimagalan@localhost ~]$
```

4. Create a play to start logstash

```
- name: Start and Enable Logstash
  become: true
  ansible.builtin.service:
    name: logstash
    state: started
    enabled: true
```

5. Run the playbook again to start logstash

```
TASK [Logstash : Start and Enable Logstash] *****
changed: [192.168.56.113]
changed: [192.168.56.114]

PLAY RECAP *****
192.168.56.113 : ok=7  changed=2  unreachable=0  failed=0  skipped=1  rescued=0  ignored=0
192.168.56.114 : ok=5  changed=1  unreachable=0  failed=0  skipped=3  rescued=0  ignored=0

qinagalan-workstation@workstation:~/ElasticStackInstaller_Galan$
```

Difficulties:

The demonstration of steps above shows two master nodes setup. This strays from the original documentation guide that demonstrates a master node and a joining node in the cluster. However due to the contradicting difficulties below, it was not accomplished.

The official documentation of elasticsearch is difficult to follow, and extremely confusing. There are two errors that elasticsearch will encounter when setting up tokens for the master node and the nodes joining the cluster:

- The joining node is already started and has its own cluster and could not join other existing cluster
 - Fix: Do not start the elasticsearch service on the joining node, only start the service for the master node
 - Result: Successful
- Second error shows that the joining node **could not join because there are security configurations implemented** already and would need to overwrite it
 - Fix tried: The play writing the configuration file for the joining node was cleared through the elasticsearch-reconfigure-node exe.

Log monitoring tools are not an optional part of system administration, but it is an imperative part of all processes and services running on a machine. Having server automation does not automatically mean letting the application or service run completely on its own. There will be scenarios where errors may suddenly appear and without having a log monitoring tool, it becomes impossible to track the causes of failure. Every debugging process also starts with identifying the errors that occurred within the process' runtime. The log monitoring tools also provide the centralized view of the activities of the system. Regardless of the specific event that occurred whether an error or

authentication problems, these are only visible due to logs

Conclusion

I have learned important concepts in managing installation of a multi-service application stack such as the ELK. Application stack shares a common GPG key to sign and authenticate packages added to the repository sources list. Without these the installation from deb and rpm packages will fail due to authentication. Another useful feature of ansible I have learned is the register variable that we can use to suppress standard output and store the value of output for usage of other plays. This is especially useful for storing the enrollment tokens that we can use to provide connections for Kibana and nodes in Elastic Stack.

Documentation Sources:

ElasticSearch Debian:

<https://www.elastic.co/docs/deploy-manage/deploy/self-managed/install-elasticsearch-with-debian-package>

ElasticSearch RedHat:

<https://www.elastic.co/docs/deploy-manage/deploy/self-managed/install-elasticsearch-with-rpm>

Logstash: <https://www.elastic.co/guide/en/logstash/7.17/installing-logstash.html>

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Claude

Purpose of Use:

To help with clarifying the confusing elasticsearch documentation

Extent of Use:

The overall structure of the playbooks and variable files were all created by me, with some guidance of ansible syntax with Claude. Hence, the unpolished structure of the playbook and variables.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.